

Design, Calibration and Assembly of 1 Liter Water Bottle using Autodesk Fusion 360

Introduction

This project presents the complete design, calibration, and assembly of a 1 liter water bottle using Autodesk Fusion 360 (Education License). The model includes a precisely calibrated bottle body, a rotating threaded cap, accurate volume control, and customized name embossing. The design closely follows a real commercial bottle to ensure dimensional and functional accuracy.

Software and Tools Used

Software Used: Autodesk Fusion 360 (Education License)

Workspace: Design (Solid Modeling and Assembly)

Major Tools: Sketch, Revolve, Extrude, Shell, Fillet, Thread, Joint, Inspect, Section Analysis, Appearance

Reference Model and Dimensions

A standard commercial 1 liter water bottle was used as a reference for calibration.

Reference specifications:

- Capacity: 1 Liter (1000 ml)
- Height: 28.5 cm
- Base Diameter: 8.2 cm

These dimensions were used as constraints while modeling to match real-world proportions.



Detailed Bottle Modeling Steps

Step 1: A new design file is created and units are set to millimeters for precision.

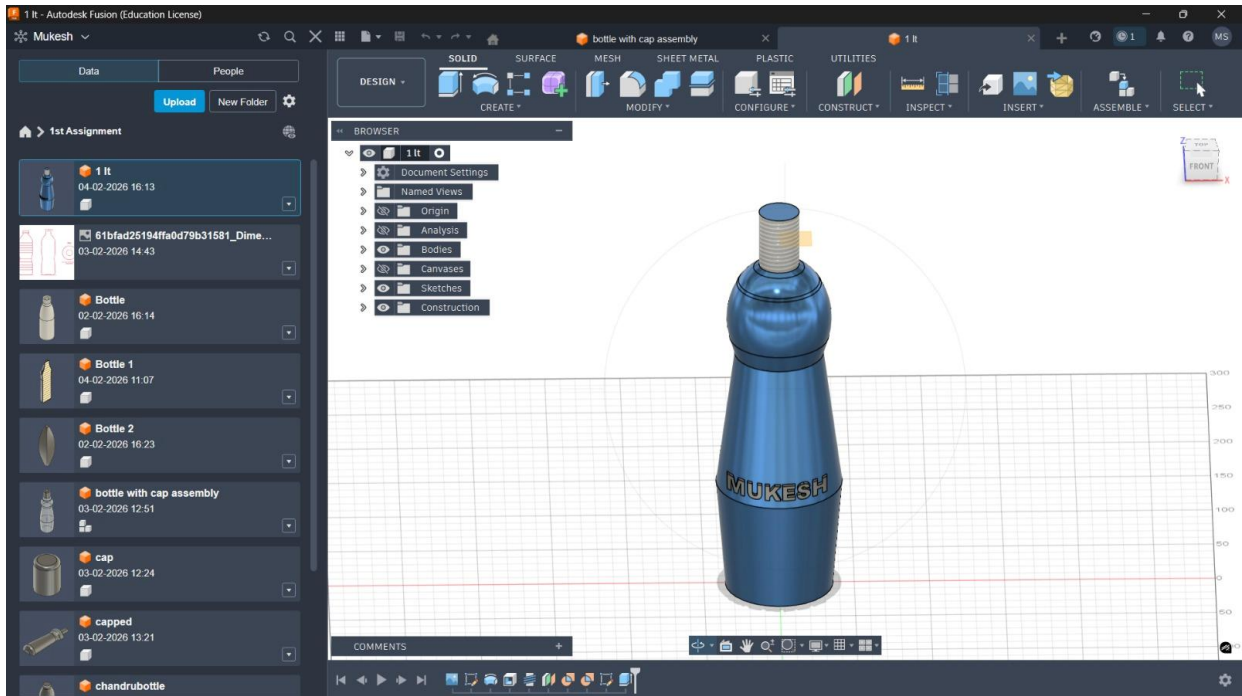
Step 2: A sketch is created on the front plane with a vertical centerline as the axis of symmetry.

Step 3: Half of the bottle profile is drawn using lines, arcs, and splines based on the reference height and diameter.

Step 4: Tangency and curvature continuity are maintained to achieve a smooth ergonomic shape.

Step 5: The Revolve command is used to create the 3D bottle body.

Step 6: Fillets are applied at the base, shoulder, and neck areas to improve strength and user comfort.

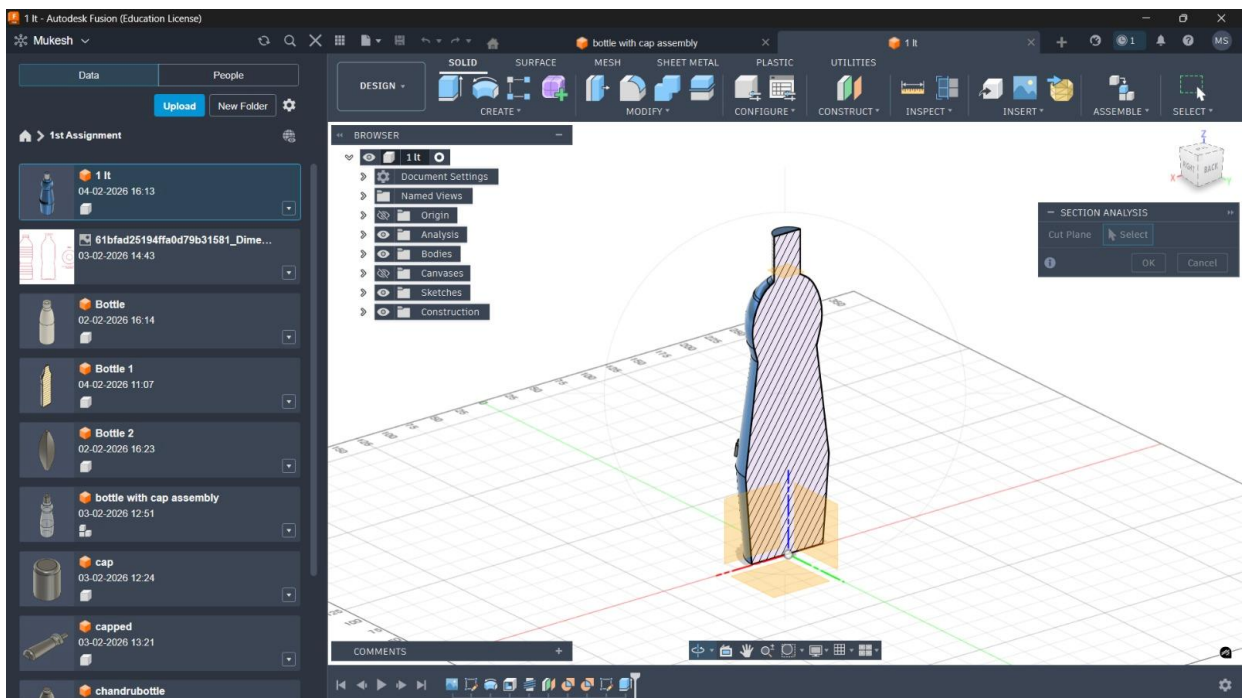


Hollowing and Wall Thickness Control

The Shell tool is applied from the top opening to hollow the bottle.

Wall thickness is kept uniform to ensure durability while minimizing material usage.

This step is crucial for achieving correct internal volume and manufacturability.



Capacity Calibration Method (1 Liter)

Calibration ensures that the bottle holds exactly 1 liter (1000 ml).

Step 1: Internal volume is checked using Inspect → Properties.

Step 2: If the volume is less than 1000 ml, the internal cavity is increased by adjusting the sketch profile or reducing wall thickness.

Step 3: If the volume exceeds 1000 ml, the body curvature or height is slightly reduced.

Step 4: This process is repeated iteratively until the volume reads 1000 ml ± tolerance.

Step 5: Section Analysis is used to visually inspect the internal cavity and confirm uniform thickness.

Bottle Neck Design and Thread Creation

The bottle neck is modeled separately within the same body.

The neck diameter and height are chosen based on standard bottle designs.

External threads are created using the Thread tool with appropriate pitch and length.

Thread calibration ensures smooth engagement with the cap.

Detailed Cap Modeling Steps

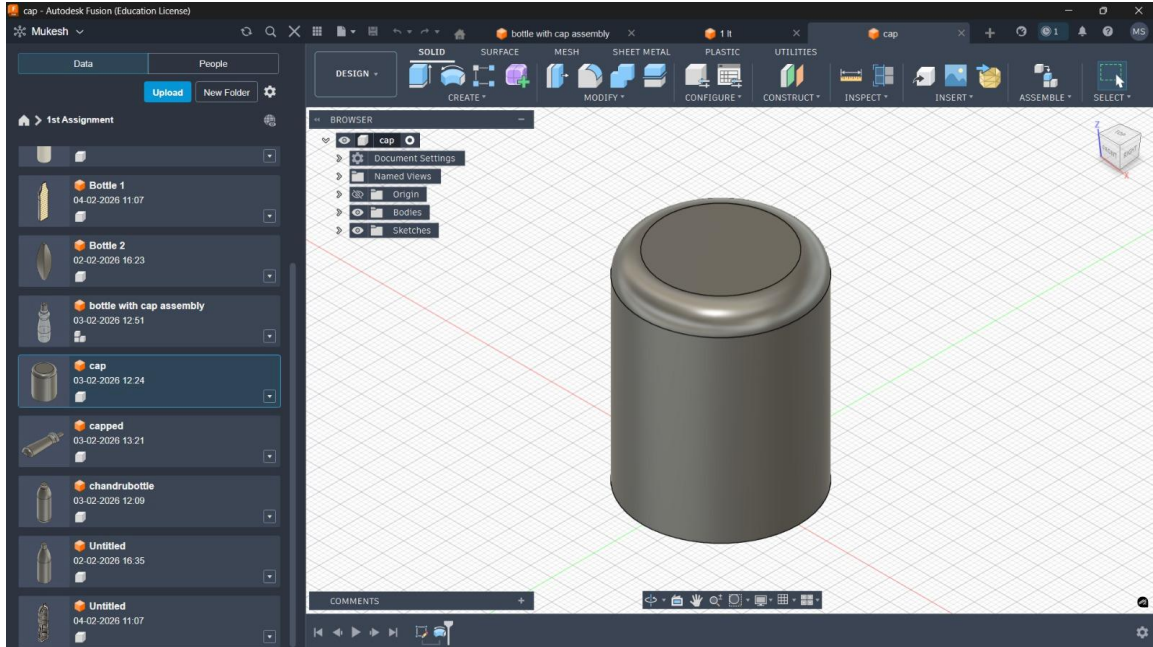
Step 1: A new component is created for the cap to allow independent motion.

Step 2: A circular sketch is drawn and extruded to form the cap body.

Step 3: The cap is hollowed using the Shell tool from the bottom face.

Step 4: Internal threads are added using the Thread command to match bottle neck threads.

Step 5: Fillets are applied to improve grip and remove sharp edges.



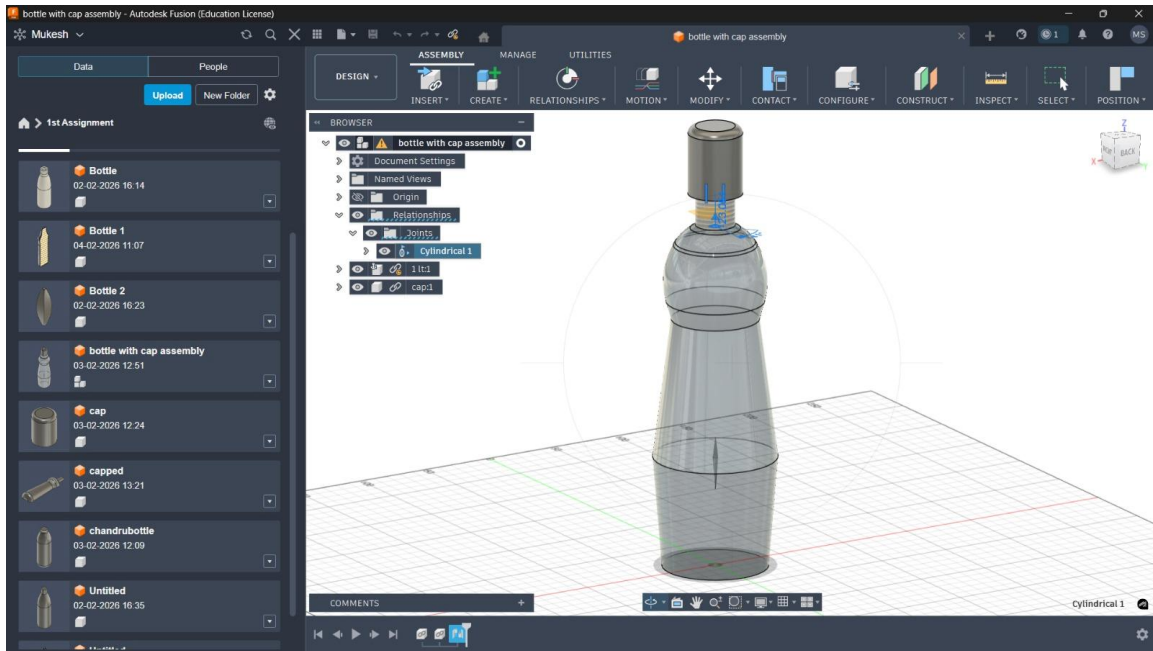
Assembly and Working Mechanism

The bottle and cap are assembled using a Cylindrical Joint.

This joint allows rotational movement while maintaining axial alignment.

Clockwise rotation closes the cap by moving it downward along the threads.

Anti-clockwise rotation opens the cap, simulating real bottle operation.



Adding Name on Bottle Surface

A tangent plane is created on the curved bottle surface.

The Text tool is used to type the name 'MUKESH'.

The text is carefully positioned and extruded outward to create an embossed effect.

This demonstrates customization and branding in product design.

Advantages of the Design

- Accurate real-world dimensions and capacity
- Realistic rotating cap mechanism
- Professional product design workflow
- Customizable branding
- Suitable for manufacturing and 3D printing

Applications

- Water and beverage containers
- CAD and product design training
- Academic assignments and projects
- Prototype development

Conclusion

The 1 liter water bottle designed in Autodesk Fusion 360 successfully demonstrates advanced CAD modeling techniques including revolve modeling, volume calibration, threaded assembly, and customization. The use of a reference bottle ensures accuracy and real-world relevance, making this project suitable for academic and industrial use.